

02464 Artificial Intelligence and Human Cognition

Bayesian Models of Visual Perception

Ill-posed problems

Conditional probability

General Formula for $P(A|B)$

$$P(A|B) = \frac{P(AB)}{P(B)}$$

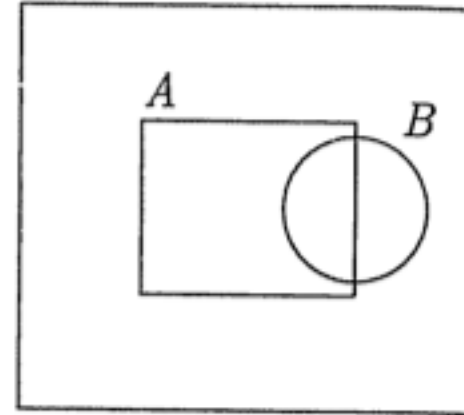
Conditional probability

Relative areas.

Suppose a point is picked uniformly at random from the big rectangle in the diagram. Imagine that information about the position of this point is revealed to you in two stages, by the answers to the following questions:

Question 1. Is the point inside the circle B ?

Question 2. Is the point inside the rectangle A ?



If the answer to Question 1 is yes, what is the probability that the answer to Question 2 will be yes?

The problem is to find the probability that the point is in the rectangle A given that it is in the circle B . By inspection of the diagram, approximately half the area inside B is inside A . So the required probability is

$$P(A|B) = \frac{P(AB)}{P(B)} = \frac{\text{Area}(AB)}{\text{Area}(B)} \approx 1/2$$

Conditional probability

General Formula for $P(A|B)$

$$P(A|B) = \frac{P(AB)}{P(B)}$$

Multiplication Rule

$$P(AB) = P(A|B)P(B)$$

$$P(BA) = P(B|A)P(A)$$

Bayes' rule

$$P(A | B)P(B) = P(AB) = P(B | A)P(A)$$

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

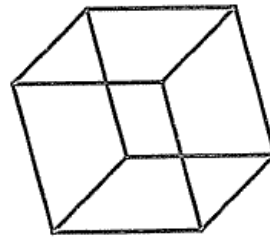
What's the fuzz?!

$$P(w \mid d) = \frac{P(d \mid w)P(w)}{P(d)}$$

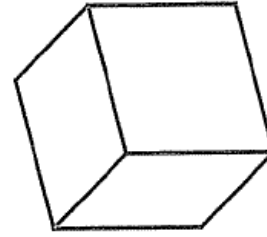
- The problem of *any* observer
- The problem of perception
 - Probability of the world given sensory *data*
- Likelihood contains a model of the world
 - Not enough information
 - Prior adds information

Bayes vs. Necker

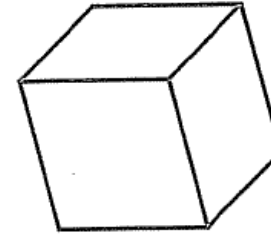
- Vision is an ill-posed problem
- To solve it, we need additional information



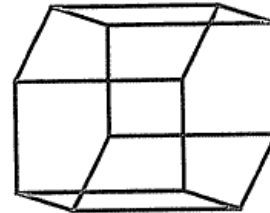
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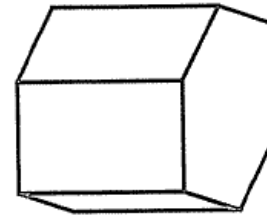
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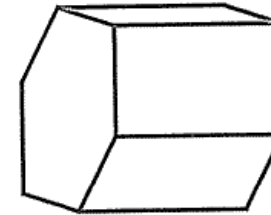
(C)



(D)



(E)



(F)

Bayes vs. Necker

- How many parameters do we need to specify the vertices in 2D?
- How many parameters do we need to specify the vertices in 3D?
- How much information is missing?
 - A simple subtraction, but...
 - Bistability means that the 3D figure is not completely specified

The (basic) Bayesian

- We get a 2D image,
- We want to infer a 3D world

$$P(w \mid d) = \frac{P(d \mid w)P(w)}{P(d)}$$

The Bayesian approach

$$P(w \mid d) = \frac{P(d \mid w)P(w)}{P(d)}$$

- Sensory data, d , is a 2-D image
- The world, w , is a 3-D structure
 - Simplification: only the 8 vertices of the cube
- Solving this would give us a probability density over all possible states of the world
- We seek the *maximum a posteriori* (MAP) solution
$$w_{map} = \arg \max_w P(w \mid d) = \arg \max_w P(d \mid w) P(w)$$
- The denominator (evidence), $P(d)$ does not depend on the world

The Maximum Likelihood Estimate (MLE)

- An uninformative, flat, prior has no influence on the solution

$$p(w | d) = \frac{p(d | w)p(w)}{p(d)} = \frac{p(d | w)\cancel{p(w)}}{\int_w p(d | w)\cancel{p(w)}}$$

- Posterior is proportional to likelihood (MAP is MLE)

$$w_{mle} = \arg \max_w p(d | w)$$

- or

$$w_{mle} = \arg \max_w P(w | d) = \arg \max_w \log P(w | d)$$

$$w_{mle} = \arg \min_w (-\log p(d | w))$$

The likelihood

- The 2-D sensory data, d , is generated by the 3-D world by
 - 2-D projection, μ , of the 3D world, w , onto the image plane
 - Adding IID Gaussian noise

$$P(d \mid w) = \prod_{n=1}^N f(d_n, \mu_n, \sigma)$$

The Maximum Likelihood Estimate (MLE)

- Take the logarithm of the likelihood

$$\log P(d | w) = \sum_{n=1}^N f(d_n, \mu_n, \sigma) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (d_n - \mu)^2$$

- Find the maximum likelihood estimate by minimising the squared error

$$w_{mle} = \arg \min_w -\log P(d | w) = \arg \min_w \sum_{n=1}^N (d_n - \mu)^2$$

The Maximum Likelihood Estimate (MLE)

- Take the logarithm of the likelihood

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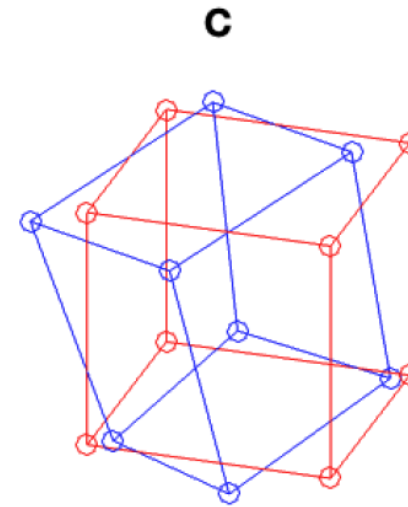
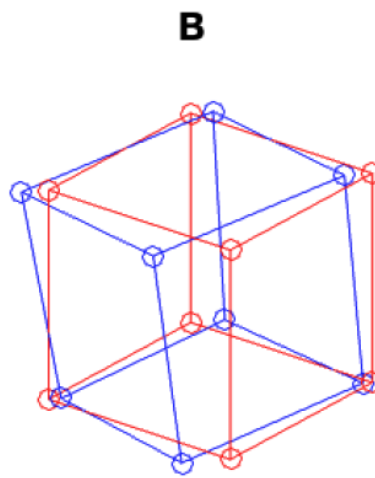
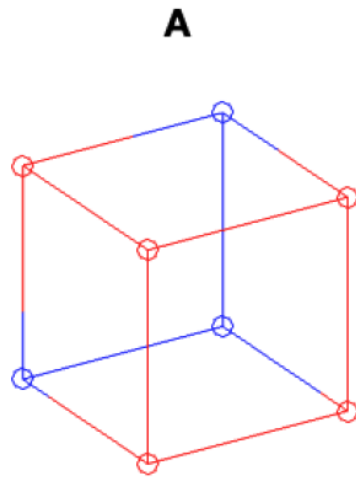
Does not depend on w

- Find the maximum likelihood estimate by minimising the squared error

$$w_{mle} = \arg \min_w -\log P(d | w) = \arg \min_w \sum_{n=1}^N (d_n - \mu)^2$$

The Maximum Likelihood Estimate (MLE)

- The MLE estimate $w_{mle} = \arg \min_w -\log P(d | w) = \arg \min_w \sum_{n=1}^N (d_n - \mu)^2$



The prior

- Gaussian prior on , $\mu_p = 90$ degree angles

$$P(w) = \prod_{m=1}^M f(a_m, \mu_p, \sigma_p)$$

The maximum a posteriori (MAP) estimate

- The logarithm of the posterior

$$\begin{aligned}\log (P (d \mid w) P (w)) &= \log P (d \mid w) + \log P(w) \\ &= \underbrace{-\frac{N}{2} \log (2 \pi \sigma^2) - \frac{1}{2 \sigma^2} \sum_{n=1}^N (d_n - \mu)^2}_{\text{Likelihood terms}} - \underbrace{\frac{M}{2} \log (2 \pi \sigma_A^2) - \frac{1}{2 \sigma_p^2} \sum_{n=1}^M (a_n - \mu_p)^2}_{\text{Prior terms}}\end{aligned}$$

The maximum a posteriori (MAP) estimate

- The logarithm of the posterior

$$\log (P (d \mid w) P (w)) = \log P (w \mid d) + \log P (w)$$

$$= -\frac{N}{2} \log (2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{n=1}^N (d_n - \mu)^2 - \frac{M}{2} \log (2\pi\sigma_p^2) - \frac{1}{2\sigma_p^2} \sum_{n=1}^M (a_n - \mu_p)^2$$

- The MAP estimate

Does not depend on w

Does not depend on w

$$w_{map} = \arg \min_w -\log P (d \mid w)$$

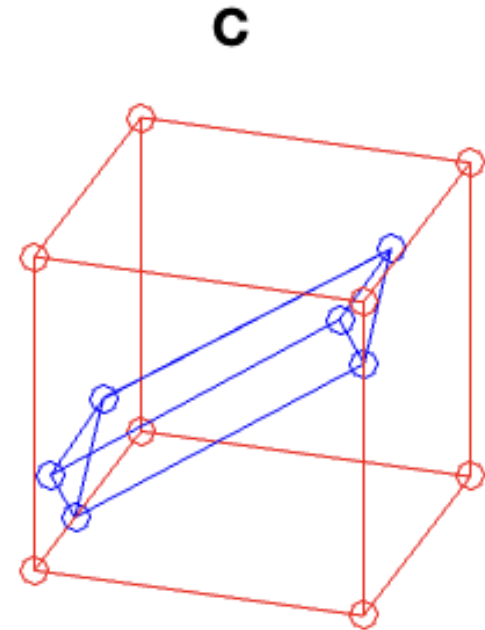
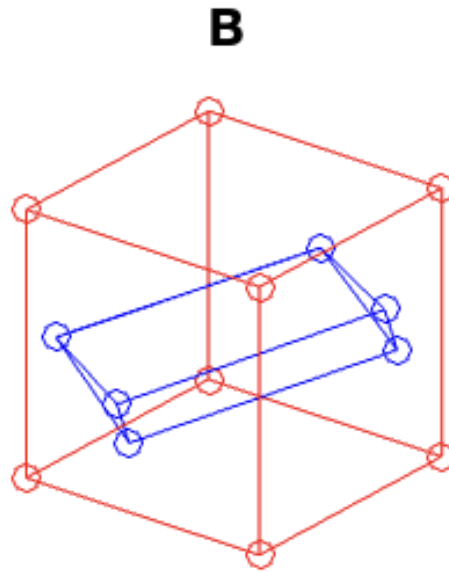
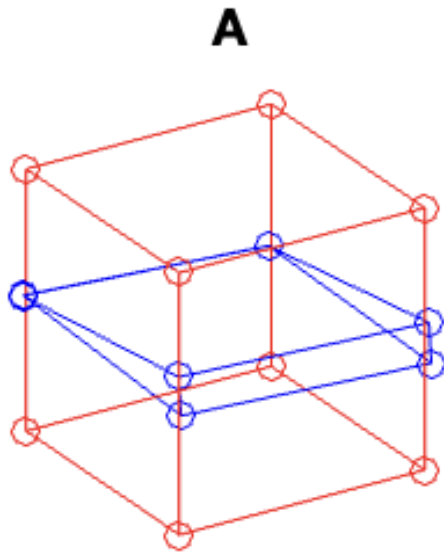
$$= \arg \min_w \left(\frac{1}{\sigma^2} \sum_{n=1}^N (d_n - \mu)^2 + \frac{1}{\sigma_p^2} \sum_{m=1}^M (a_m - \mu_p)^2 \right)$$

The maximum a posteriori (MAP) estimate

$$w_{map} = \arg \min_w \left(\sum_{n=1}^N (d_n - \mu)^2 + \frac{r_p}{r} \sum_{m=1}^M (a_m - \mu_p)^2 \right)$$

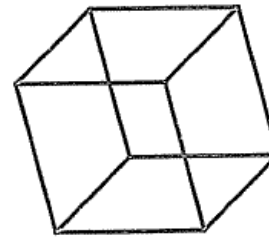
- The MAP estimate is a *weighted* sum of squares
- The weight is the ratio of sensory and prior reliability
 - Sensory reliability is related to sensory noise
 - Prior reliability is related to how informative the prior is
 - If the weight, r_p / r , is too small then the prior has no effect
 - MAP estimates similar to MLE estimates
 - If the weight is too large then the sensory data in the likelihood has no effect
 - MAP estimates has right angles but do not look like a cube from any angle

The maximum a posteriori (MAP) estimate

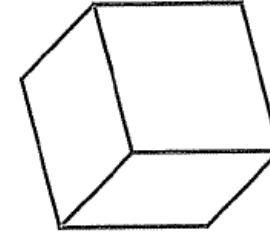


Bayes vs. Necker

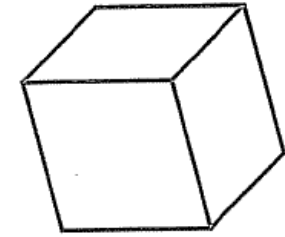
- Implementing Bayes rule for the Necker cube requires additional assumptions
- These assumptions should generalize to other problems
 - 90 deg prior does not
- Testable hypotheses



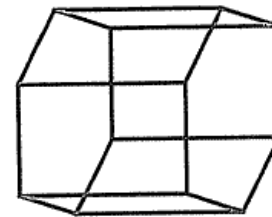
(A)



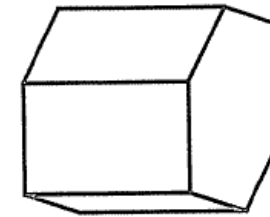
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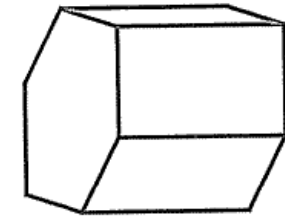
(C)



(D)



(E)



(F)

Exercise

- Find the Matlab (sorry) files on DTU Learn
 - Insert the expression for the log posterior (one line of code)
 - Try the MLE solution for Necker
 - Try a prior (angles) for Necker
 - Try the same for the hexagonal prism
 - Can you find a prior that works for both?